

Part I

SUMMARY of Derivatives as Functions

Summary

February 21, 2026

Summary

- The derivative of a function f is a function f' whose domain consists of all points in the domain of f where the function f is differentiable. The values of f' are given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

where this value exists.

- If f is differentiable at $x = a$, then f is continuous at $x = a$.

Part II

Recitation Questions

Problem 1

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Problem 1

Problem 1 (a) Suppose $f'(2)$ exists. Which of the following must be true?

- (i) $\lim_{x \rightarrow 2} f(x)$ must exist, but $\lim_{x \rightarrow 2} f(x) \neq f(2)$
- (ii) $\lim_{x \rightarrow 2} f(x) = f(2)$.
- (iii) $\lim_{x \rightarrow 2} f(x) = f'(2)$
- (iv) $\lim_{x \rightarrow 2} f(x)$ need not exist.

(b) Assuming that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$, which of the following is true?

- (i) $\frac{0}{0} = 1$
- (ii) the tangent line to $y = \sin(x)$ at $(0, 0)$ has slope 1.
- (iii) you can cancel the x 's.
- (iv) for all x near 0, $\sin(x) = x$.
- (v) for all x near 0, $\sin(x) \approx x$.

Problem 2

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Problem 2

(a) *Fill in the blanks*

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

(b) *Let*

$$f(x) = \frac{1}{x+4}.$$

Use the (limit) definition of derivative in (a) to find $f'(x)$.

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Problem 3

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Problem 3

Problem 3 Let $f(x) = |5 - x|$.

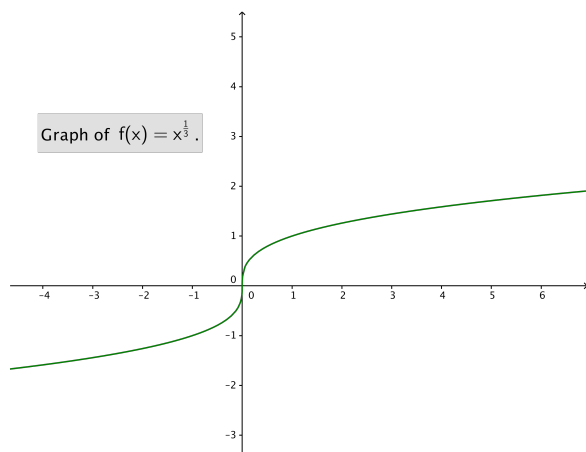
- (a) For $a < 5$, find $f'(a)$.
- (b) For $a > 5$, find $f'(a)$.
- (c) Determine whether $f'(5)$ exists.
- (d) Sketch a graph of the function $f(x)$ and its derivative $f'(x)$

Problem 4

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Problem 4

Problem 4 Define the function f by $f(x) = x^{1/3}$ and consider the graph of this function:



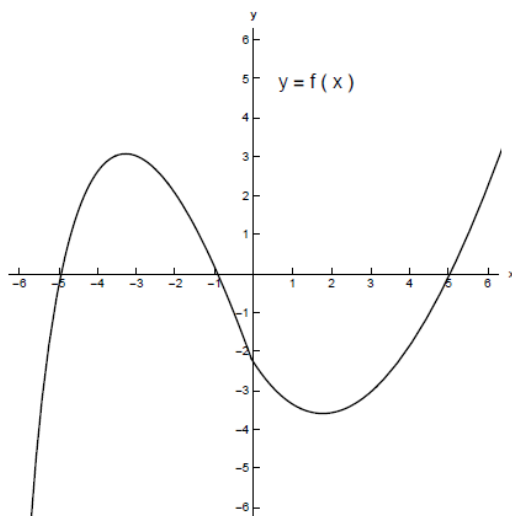
Which of the following two statements are true:

- (a) The graph of f has a tangent line at $x = 0$.
- (b) The derivative $f'(0)$ is defined.

Problem 5

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Problem 5 Suppose we are given the graph of a function f :



- (a) Use this graph to find the following: (Assume all values will be integers or $+\infty$ or $-\infty$)
- (i) all x where $f(x) = 0$,
 - (ii) all x where $f(x) > 0$,
 - (iii) all x where $f(x) < 0$, and
 - (iv) all x where $f(x)$ attains a local maximum and all x where f attains a local minimum.

Without sketching the graph of f' find

- (b) (i) all x where $f'(x) = 0$,
- (ii) all x where $f'(x) > 0$,
- (iii) all x where $f'(x) < 0$, and
- (iv) On the following intervals, does $f'(x)$ seem to be increasing or decreasing?
- i. $(-\infty, 0)$
 - ii. $(0, \infty)$

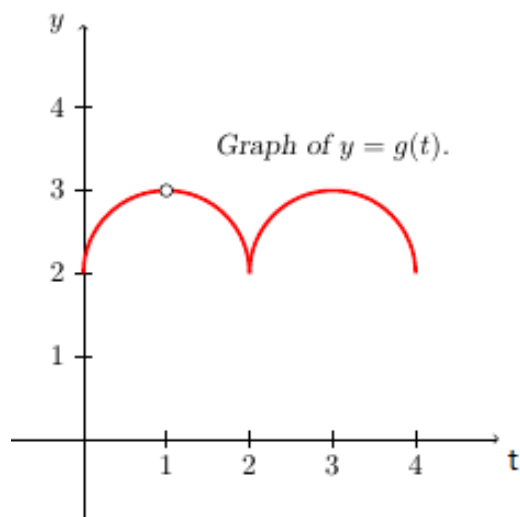
- (c) Sketch a graph of f' .

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Problem 6

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Problem 6 Use the graph of g



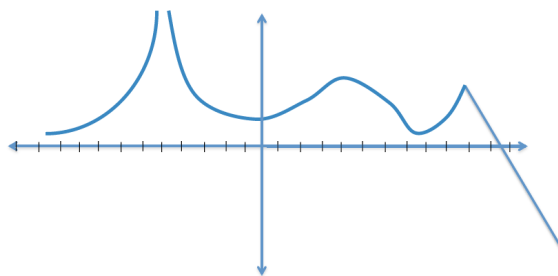
- (a) Find the values of t in $(0, 4)$ at which g is not continuous.
- (b) Find the values of t in $(0, 4)$ at which g is not differentiable.

Problem 7

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Problem 7

Problem 7 Given the following graph of a function h sketch a graph of the derivative h' .



[2]